E28: the GLS fold on E19′s single-τ_T line (stages 0, A, A2)

Run report and results, 2026-10-30. Note `RESEARCH_GLV_INVARIANT_FACTOR_BASES.md` §9.26 (pre-registration, `4845b32c7`) and §9.30 (results). Binary `789dc2ff6` (Stage 0, Stage A); supplementary control `0e4fccce1`. Records `experiments/33_tau_stage_a/`. Host: the 4-vCPU container, no isolation tier (isolab-eks unreachable); operation counts only. **No Stage B was run** (by instruction).

What ran

- Stage 0: 40 GLS twists over $\mathbb{F}_{p^4}$ on E19′s line ($\sigma(x_0) = -x_0, c \in \mathbb{F}_p$), $p \in \{53, 101, 149, 197, 269\}$, seeds 1–8. 0 failures.
- Stage A: 10 curves (seeds 1, 2): one exact-oracle stream (`mitm_signed`, every decomposition re-added), two arms on the same points: τ (E19′s fold) and τψ ($\langle \tau_T, \psi \rangle$); Wiedemann to the verified log; 16 $\langle \pm\psi \rangle$ and 16 negation walks per curve in E25′s and E27′s arithmetic (640 walks, all correct).
- Solve-cost control: E19′s weighted D_4 solve on 3 uniform targets per curve (registered; vacuous agreement, no target had a decomposition) and, as a recorded deviation, on 3 targets per curve the oracle decomposes (30 calls).

Measured

- Stage 0: every ψ and line check holds on all 40 curves; $\boldsymbol{h = 4}$ **on all 40**; density 0.50–0.52; 4 / 8 points a column (with ±).
- Stage A: columns ratio 2.00; decompositions per target median 8; residuals to the log pooled 40,749 / 23,345 (×1.75; per curve 1.36–2.56); y pooled 98.4 / 110.1; every log recovered.
- Control: C median $1.36–1.44 \cdot 10^9$ (uniform), $1.37–1.48 \cdot 10^9$ (decomposed); all 30 decomposed calls equal the oracle's set.
- Stage A2 (**model**, derived from measured constants): τψ arm $6.99 \cdot 10^5–1.74 \cdot 10^7$ × the $\langle \pm\psi \rangle$ walk at 0 GAE; ×1.30–2.82 below the τ arm on the same curve; against E19 at the same p: by size ×1.77–4.29.

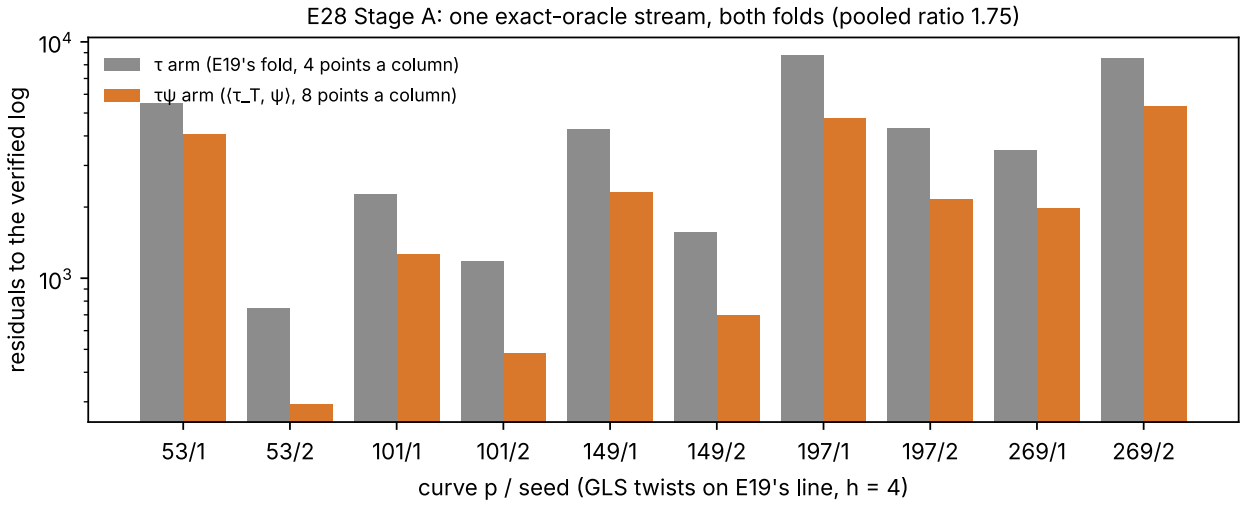


Figure 1: Residuals to the verified logarithm per arm (Stage A). Source: `stage_a_p*s*.json`, binary `789dc2ff6`.

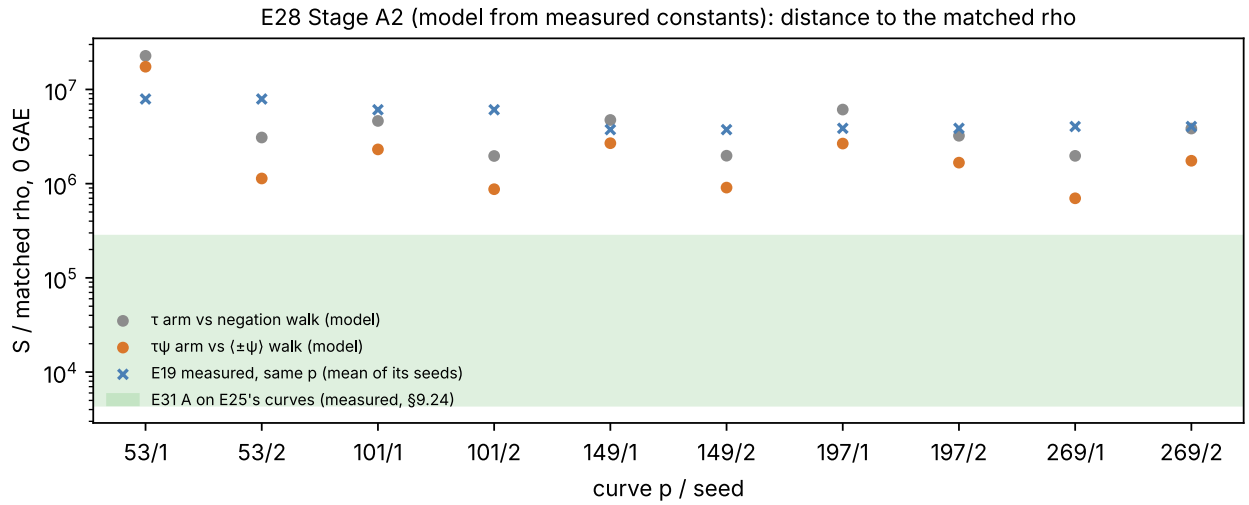


Figure 2: Stage A2 model distances against matched walks, E19′s measured rows and E31 A's measured range. Source: `summary.json`.

Verdicts (§9.26)

G2.100 holds (gate passed). G2.101 holds (pooled). G2.102 holds with the recorded deviation. G2.103 falsified per curve and at $p = 101$ by size; the $\geq 10\times$ gap to E31 A fails at $p = 149$ under the strict reading (best E28 row vs worst E31 row, 3.3×), holds at the other matched sizes.

What it says

The GLS fold carries over to the single-τ_T line with $h = 4$ (unlike the V_4 line, §9.17) and halves its columns; with E19′s unchanged solve the line's modelled cost moves $\approx$ 2–4× against its own matched walks. **Engineering on the D_4 line, a model, not an improvement of the baseline**; distance remaining $\geq 7 \cdot 10^5\times$ (E25 arithmetic). No leaderboard or `benchmarks/` row: no measured end-to-end S and no validation-gate evidence.